

M2 de Mathématiques Fondamentales

Algebraic Number Theory I

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Chapter 2

Valued fields

2.1 Absolute values

2.1.1 Definitions

In the previous chapter, we have seen that, given a number field K and a non-zero prime ideal $\mathfrak{p}$ in $\mathcal{O}_K$, one can endow K with a $\mathfrak{p}$ -adic valuation $v_{\mathfrak{p}} : K^{\times} \rightarrow \mathbb{Z}$. The properties satisfied by $v_{\mathfrak{p}}$ motivate the following definition:

Definition 2.1.1

Let K be a field. A **valuation** on K is a map $v : K \rightarrow \mathbb{R} \cup \{\infty\}$ such that:

1. $\forall x \in K, v(x) = \infty \Leftrightarrow x = 0$;
2. $\forall x, y \in K, v(xy) = v(x) + v(y)$;
3. $\forall x, y \in K, v(x + y) \geq \min\{v(x), v(y)\}$.

It is said to be **discrete** if $v(K^{\times}) = a\mathbb{Z}$ for some $a \in \mathbb{R}_{>0}$. An element $\pi \in K^{\times}$ with $v(\pi) = a$ is called a **uniformizer**. If $a = 1$, the valuation is called **normalized**.

The main goal of this Chapter consists in studying fields endowed with a valuation, or more generally an absolute value:

Definition 2.1.2

A **valued field** is a field K endowed with an **absolute value** $|\cdot|$, that is a map $|\cdot| : K \rightarrow \mathbb{R}_{\geq 0}$ such that:

1. $\forall x \in K, |x| = 0 \Leftrightarrow x = 0$;
2. $\forall x, y \in K, |xy| = |x||y|$;
3. $\forall x, y \in K, |x + y| \leq |x| + |y|$.

The field K (or the absolute value $|\cdot|$) is said to be **nonarchimedean** if:

$$\forall x, y \in K, |x + y| \leq \max\{|x|, |y|\}.$$

It is said to be **archimedean** otherwise.

Example 2.1.3.

1. Given a field K , the **trivial** absolute value on K is defined by $|x|_0 = 1$ for all $x \in K^\times$. **In the rest of this course, we will always tacitly assume that absolute values are non-trivial.**
2. If K is a number field and $\sigma \in \text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$, one defines an archimedean absolute value on K by the formula $|x|_\sigma = |\sigma(x)|$.
3. Let K be a field. There is a bijection between valuations on K and nonarchimedean absolute values on K :

$$\begin{aligned} \{(\text{Non-trivial}) \text{ valuations}\} &\rightarrow \{\text{nonarchimedean absolute values}\} \\ v &\mapsto (|\cdot|_v : x \mapsto e^{-v(x)}) \\ (v_{|\cdot|} : x \mapsto -\ln|x|) &\leftarrow |\cdot|. \end{aligned}$$

In particular, if K is a number field and $\mathfrak{p}$ is a non-zero prime ideal in $\mathcal{O}_K$, one defines an absolute value on K by the formula $|x|_{v_{\mathfrak{p}}} = e^{-v_{\mathfrak{p}}(x)}$.

Proposition 2.1.4

Let $(K, |\cdot|)$ be a valued field. The following assertions are equivalent:

- (i) the absolute value $|\cdot|$ is nonarchimedean;
- (ii) the sequence $(|n|)_{n \geq 0}$ stays bounded by 1;
- (iii) the sequence $(|n|)_{n \geq 0}$ stays bounded.

In particular, an archimedean valued field has characteristic 0.

Proof. If K is nonarchimedean, then, for $n \geq 0$:

$$|n| = |1 + \cdots + 1| \leq \max\{|1|, \dots, |1|\} = 1.$$

Hence (i) implies (ii). Of course, (ii) implies (iii). It remains to prove (iii) implies (i). For that purpose, assume that there exists $M > 0$ such that $|n| \leq M$ for all $n \geq 0$. Take $x, y \in K$. Then, for all $n \geq 1$:

$$|x + y|^n \leq \sum_{i=0}^n \binom{n}{i} |x|^i |y|^{n-i} \leq M(n+1) \max\{|x|, |y|\}^n,$$

so that:

$$|x + y| \leq \sqrt[n]{M(n+1)} \max\{|x|, |y|\} \xrightarrow{n \rightarrow \infty} \max\{|x|, |y|\}.$$

We deduce that K is nonarchimedean. □

2.1.2 Topology on a valued field

If $(K, |\cdot|)$ is a valued field, one can define a distance on K by setting $d(x, y) := |x - y|$. This allows to endow K with the structure of a topological field.

Definition 2.1.5

Let K be a field. Two absolute values on K are said to be **equivalent** if they define the same topology on K . An equivalence class of absolute values on K is called a **place** of K .

Proposition 2.1.6

Let K be a field and let $|\cdot|$ and $|\cdot|'$ be two absolute values on K . The following assertions are equivalent:

- (i) the absolute values $|\cdot|$ and $|\cdot|'$ are equivalent;
- (ii) for every $x \in K$, we have $|x| < 1$ if, and only if, $|x|' < 1$;
- (iii) for every $x \in K$, if $|x| < 1$, then $|x|' < 1$;
- (iv) there exists $s > 0$ such that $|\cdot|' = |\cdot|^s$.

Proof. Assume (i). For $x \in K$, we have the equivalences:

$$|x| < 1 \Leftrightarrow x^n \xrightarrow[n \rightarrow \infty]{|\cdot|} 0 \Leftrightarrow x^n \xrightarrow[n \rightarrow \infty]{|\cdot|'} 0 \Leftrightarrow |x|' < 1,$$

hence (ii).

Now assume (iii). Fix some $y \in K$ with $|y| > 1$ (such an element exists because $|\cdot|$ is not trivial), and take some $x \in K^\times$. Take $\alpha \in \mathbb{R}$ such that $|x| = |y|^\alpha$ and consider $(p_n/q_n)_{n \geq 0}$ a sequence of rational numbers converging to α from above and $(r_n/s_n)_{n \geq 0}$ a sequence of rational numbers converging to α from below. Then:

$$|y|^{r_n/s_n} < |x| = |y|^\alpha < |y|^{p_n/q_n},$$

so that:

$$|y^{r_n} x^{-s_n}| < 1 \quad \text{and} \quad |x^{q_n} y^{-p_n}| < 1.$$

We deduce that:

$$|y^{r_n} x^{-s_n}|' < 1 \quad \text{and} \quad |x^{q_n} y^{-p_n}|' < 1,$$

so that:

$$|y|^{r_n/s_n} < |x|' < |y|^{p_n/q_n}.$$

Making n go to infinity, we get $|x|' = |y|'^\alpha$. This being true for all $x \in K^\times$, we deduce that $|x|' = |x|^s$ with $s = \frac{\ln |y|'}{\ln |y|}$, hence (iv).

The implications (ii) $\Rightarrow$ (iii) and (iv) $\Rightarrow$ (i) are obvious. $\square$

Corollary 2.1.7

Let $|\cdot|_1, \dots, |\cdot|_n$ be pairwise non-equivalent absolute values on a field K . There exists $z \in K$ such that $|z|_1 > 1$ and $|z|_i < 1$ for $2 \leq i \leq n$.

Proof. We proceed by induction on n . For $n = 1$, non-triviality gives an element of absolute value greater than one. For $n = 2$, since $|\cdot|_1$ and $|\cdot|_2$ are not equivalent, there exists $a \in K$ such that $|a|_1 < 1$ and $|a|_2 \geq 1$ and $b \in K$ such that $|b|_1 \geq 1$ and $|b|_2 < 1$. Setting $x := b/a$, we get $|x|_1 > 1$ and $|x|_2 < 1$, as wished.

Assume now that $n \geq 3$. By the case $n = 2$, we can find x such that $|x|_1 > 1$ and $|x|_n < 1$, and by the inductive assumption, we can find $y \in K$ such that $|y|_1 > 1$ and $|y|_i < 1$ for $2 \leq i \leq n-1$. If $|y|_n \leq 1$, we may take $z = xy^r$ with r large enough. If $|y|_n > 1$, we may take $z = x \frac{y^r}{1+y^r}$ with r large enough. $\square$

Theorem 2.1.8 (Artin-Whaples Weak Approximation)

Let $|\cdot|_1, \dots, |\cdot|_n$ be pairwise non-equivalent absolute values on a field K . Consider the product $\prod_{i=1}^n K$ where the i -th term is endowed with the

topology induced by $|\cdot|_i$. The image of the diagonal map

$$\begin{aligned}\delta : K &\rightarrow \prod_{i=1}^n K, \\ a &\mapsto (a, \dots, a)\end{aligned}$$

is dense.

Proof. By the previous Corollary, for each $1 \leq i \leq n$, we can find z_i such that $|z_i|_i > 1$ and $|z_i|_j < 1$ if $j \neq i$. Then

$$\frac{z_i^m}{1 + z_i^m} \xrightarrow{m \rightarrow \infty} \begin{cases} 1 & \text{for } |\cdot|_i \\ 0 & \text{for } |\cdot|_j, j \neq i, \end{cases}$$

hence for each $(a_1, \dots, a_n) \in \prod_{i=1}^n K$:

$$\delta \left(\sum_{i=1}^n \frac{z_i^m}{1 + z_i^m} a_i \right) \xrightarrow{m \rightarrow \infty} (a_1, \dots, a_n) \in \prod_{i=1}^n K.$$

□



Emil Artin (Vienna 1898 - Hamburg 1962)

*No picture seems to be available for George William Whaples
(Neponset 1914 - Amherst 1981)*

2.1.3 Places of $\mathbb{Q}$

In this subsection, we classify all places of the field of rational numbers.

Theorem 2.1.9 (Ostrowski, 1916)

A system of representatives of the set of places of $\mathbb{Q}$ is given by the usual absolute value $|\cdot|_\infty$ and by the p -adic absolute values $|\cdot|_{v_p}$ for p a prime number.



Alexander Ostrowski (Kiev 1893 - Lugano 1986)

Proof. Let $|\cdot|$ be a nonarchimedean absolute value on $\mathbb{Q}$. By Proposition 2.1.4, we know that $|n| \leq 1$ for all $n \in \mathbb{Z}$. Set $\mathfrak{p} := \{n \in \mathbb{Z} : |n| < 1\}$. Since $|\cdot|$ is nontrivial, $\mathfrak{p} \neq \{0\}$. Moreover:

- for $m, n \in \mathfrak{p}$, we have $|m + n| \leq \max\{|m|, |n|\} < 1$, hence $m + n \in \mathfrak{p}$;
- for $a \in \mathbb{Z}$ and $n \in \mathfrak{p}$, we have $|an| = |a||n| < 1$, hence $an \in \mathfrak{p}$;
- for $a, b \in \mathbb{Z}$ such that $ab \in \mathfrak{p}$, we have $|ab| < 1$, $|a| \leq 1$ and $|b| \leq 1$, hence $|a| < 1$ or $|b| < 1$, and $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$.

We deduce that $\mathfrak{p}$ is a non-zero prime ideal in $\mathbb{Z}$. Write $\mathfrak{p} = p\mathbb{Z}$ with p a prime number. Given $a \in \mathbb{Z} \setminus \{0\}$, we have $a = bp^{v_p(a)}$ for some $b \in \mathbb{Z}$ coprime to p , and then $|a| = |b| \cdot |p|^{v_p(a)} = |p|^{v_p(a)} = |a|_{v_p}^s$ with $s = -\ln|p|$. Hence $|\cdot|$ is equivalent to $|\cdot|_{v_p}$. Note also that, if p and q are two distinct prime numbers, we have $|p|_{v_p} < 1$ and $|p|_{v_q} = 1$, hence $|\cdot|_{v_p}$ and $|\cdot|_{v_q}$ are not equivalent.

Let now $|\cdot|$ be an archimedean absolute value on $\mathbb{Q}$. Given $m, n > 1$, we claim that $|m| = \max\{1, |n|\}^{\log_n m}$. Indeed, write $m = a_0 + a_1 n + \dots + a_r n^r$ with $a_0, \dots, a_r \in$

$\{0, \dots, n-1\}$ and $n^r \leq m$. Then:

$$\begin{aligned} |m| &\leq \sum_{i=0}^r |a_i| \cdot |n|^i \leq \sum_{i=0}^r |a_i| \cdot 1 \cdot |n|^i \leq (r+1)n \max\{1, |n|\}^r \\ &\leq (\log_n m + 1)n \cdot \max\{1, |n|\}^{\log_n m}. \end{aligned}$$

Replacing m by m^k with $k \geq 1$, we get:

$$|m|^k \leq (k \log_n m + 1)n \cdot \max\{1, |n|\}^{k \log_n m},$$

and hence

$$|m| \leq ((k \log_n m + 1)n)^{1/k} \cdot \max\{1, |n|\}^{\log_n m} \xrightarrow[k \rightarrow \infty]{} \max\{1, |n|\}^{\log_n m}.$$

We deduce that $|m| \leq \max\{1, |n|\}^{\log_n m}$. Now recall that, by Proposition 2.1.4, there exists $n_0 \in \mathbb{N}$ such that $|n_0| > 1$. We then get:

$$1 < |n_0| \leq \max\{1, |m|\}^{\log_m n_0} = |m|^{\log_m n_0} \quad \text{and} \quad |m| \leq |n_0|^{\log_{n_0} m},$$

so that $|m| = |n_0|^{\log_{n_0} m}$. Choosing $s > 0$ such that $|n_0| = |n_0|_\infty^s$, we get:

$$|m| = |n_0|^{s \log_{n_0} m} = m^s.$$

This being true for all $m > 1$, we deduce that $|\cdot| = |\cdot|_\infty^s$. □

2.1.4 Algebraic properties of valued fields

We finish our section on absolute values by studying the algebraic properties of non-archimedean valued fields.

Proposition 2.1.10

Let $(K, |\cdot|)$ be a nonarchimedean valued field. Let v be the associated valuation.

1. The subset:

$$\mathcal{O} := \{x \in K : |x| \leq 1\} = \{x \in K : v(x) \geq 0\}$$

is a local integrally closed domain called the **valuation ring** and with maximal ideal:

$$\mathfrak{p} := \{x \in K : |x| < 1\} = \{x \in K : v(x) > 0\}.$$

2. Assume moreover that v is discrete normalized and that π is a uniformizer. Then every element $x \in K^\times$ can be written uniquely as $u\pi^m$ with $u \in \mathcal{O}^\times$ and $m \in \mathbb{Z}$ and the ring $\mathcal{O}$ is a principal ideal domain whose nonzero ideals are given by $\mathfrak{p}^m = (\pi^m)$ for $m \in \mathbb{Z}_{\geq 0}$.

Proof. 1. Except for the fact that $\mathcal{O}$ is integrally closed, this is a straightforward verification thanks to the definition of a nonarchimedean absolute value. Now take $x \in K$ satisfying an equation of the form:

$$x^d + a_{d-1}x^{d-1} + \cdots + a_0 = 0$$

with $a_0, \dots, a_{d-1} \in \mathcal{O}$. If $x \notin \mathcal{O}$, then $|x| > 1$ and hence $|x^d| = |a_{d-1}x^{d-1} + \cdots + a_0| > \max\{|a_i x^i| : 0 \leq i \leq d-1\}$: contradiction! We deduce that $x \in \mathcal{O}$ and that $\mathcal{O}$ is integrally closed.

2. Given $x \in K^\times$, we have $v(x\pi^{-v(x)}) = 0$, hence $x\pi^{-v(x)} \in \mathcal{O}^\times$ and we can write $x = u\pi^m$ with $u \in \mathcal{O}^\times$ and $m = v(x)$. The uniqueness comes from the observation that, if $x = u\pi^m$ with $u \in \mathcal{O}^\times$, then $m = v(x)$ and $u = x\pi^{-v(x)}$.

Let now I be a non-zero ideal on $\mathcal{O}$. Let $x \in I \setminus \{0\}$ such that $v(x)$ is minimal. For any $y \in I$, we have $v(x) \leq v(y)$, so that $v(yx^{-1}) \geq 0$ and $yx^{-1} \in \mathcal{O}$. We deduce that $y \in (x)$, and hence that $I = (x)$. If $x = u\pi^m$ with $u \in \mathcal{O}^\times$, then $I = (\pi^m)$.

□

2.2 Completions

2.2.1 The formal construction

Let $(K, |\cdot|)$ be a valued field. Let R be the ring of all Cauchy sequences with values in K and let $\mathfrak{m}$ be the ideal in R of all sequences in K that converge to 0.

Lemma 2.2.1. *The ideal $\mathfrak{m}$ is maximal in R .*

Proof. Take some $(a_n)_n \in R \setminus \mathfrak{m}$. There exists some $\epsilon > 0$ and some $N > 0$ such that $|a_n| > \epsilon$ for $n \geq N$. In particular, we can find some sequence $(b_n)_n$ with values in K such that $a_n b_n = 1$ for $n \geq N$. Since $(a_n)_n$ is a Cauchy sequence and $|a_n| > \epsilon$ for $n \geq N$, the sequence $(b_n)_n$ is also Cauchy. Hence $(b_n)_n \in R$ and $(a_n b_n) \equiv 1 \pmod{\mathfrak{m}}$. This shows that $R/\mathfrak{m}$ is a field. □

Definition 2.2.2

The **completion** of $(K, |\cdot|)$ is defined as $\hat{K} := R/\mathfrak{m}$. The field K is embedded in $\hat{K}$ via the map that sends $a \in K$ to the constant sequence equal to a .

Given $(a_n)_n \in R$, we have $||a_n| - |a_m|| \leq |a_n - a_m|$ for all m and n , and hence $(|a_n|)_n$ is a Cauchy sequence of real numbers. It therefore has a limit, that we denote $|(a_n)_n|$. Moreover, if $(b_n)_n \in R$ is such that $(a_n - b_n)_n \in \mathfrak{m}$, then $||a_n| - |b_n|| \leq |a_n - b_n| \xrightarrow{n \rightarrow \infty} 0$, and hence $|(a_n)_n| = |(b_n)_n|$. This shows that the formula $(a_n)_n \mapsto |(a_n)_n|$ extends the absolute value $|\cdot|$ on K to $\hat{K}$. In particular, $\hat{K}$ inherits a topological field structure such that K is dense in $\hat{K}$.

Lemma 2.2.3. *The valued field $(\hat{K}, |\cdot|)$ is complete.*

Proof. Take $(a^{(m)})_m = (a_n^{(m)})_{n,m}$ a Cauchy sequence in $\hat{K}$. Each of the sequences $a^{(m)}$ is Cauchy, hence we can find $N_m > 0$ such that $|a_n^{(m)} - a_{n'}^{(m)}| < \frac{1}{m}$ for $n, n' \geq N_m$. We set $b_m = a_{N_m}^{(m)}$ for each m .

Let $\epsilon > 0$. Since $(a^{(m)})_m$ is Cauchy, there exists $M > 0$ such that $|a^{(m)} - a^{(m')}| = \lim_{n \rightarrow \infty} |a_n^{(m)} - a_n^{(m')}| < \epsilon$ for $m, m' \geq M$. Hence, for $m, m' \geq \max\{M, \frac{1}{\epsilon}\}$ and $n \geq \max\{N_m, N_{m'}\}$:

$$\begin{aligned} |b_m - b_{m'}| &= |a_{N_m}^{(m)} - a_{N_{m'}}^{(m')}| \leq |a_{N_m}^{(m)} - a_n^{(m)}| + |a_n^{(m)} - a_n^{(m')}| + |a_n^{(m')} - a_{N_{m'}}^{(m')}| \\ &\leq \frac{1}{m} + \frac{1}{m'} + |a_n^{(m)} - a_n^{(m')}| \\ &\leq 2\epsilon + |a_n^{(m)} - a_n^{(m')}|. \end{aligned}$$

Passing to the limit as $n \rightarrow \infty$, we get $|b_m - b_{m'}| < 3\epsilon$. Hence $b := (b_m)_m$ is a Cauchy sequence and can be seen as an element of $\hat{K}$.

Now for $m \geq \max\{M, \frac{1}{\epsilon}\}$, $n \geq \max\{M, N_m, \frac{1}{\epsilon}\}$ and $r \geq \max\{N_m, N_n\}$, we have:

$$\begin{aligned} |a_n^{(m)} - b_n| &= |a_n^{(m)} - a_{N_n}^{(n)}| \\ &\leq |a_n^{(m)} - a_r^{(m)}| + |a_r^{(m)} - a_r^{(n)}| + |a_r^{(n)} - a_{N_n}^{(n)}| \\ &\leq \frac{1}{m} + \frac{1}{n} + |a_r^{(m)} - a_r^{(n)}| \\ &\leq 2\epsilon + |a_r^{(m)} - a_r^{(n)}|. \end{aligned}$$

Passing to the limit as $r \rightarrow \infty$, we get $|a_n^{(m)} - b_n| \leq 3\epsilon$. Hence, for $m \geq \max\{M, \frac{1}{\epsilon}\}$:

$$|a^{(m)} - b| = \lim_{n \rightarrow \infty} |a_n^{(m)} - b_n| \leq 3\epsilon.$$

We deduce that $a^{(m)} \xrightarrow{m \rightarrow \infty} b$. □

Remark 2.2.4. The completion $(\hat{K}, |\cdot|)$ of $(K, |\cdot|)$ turns out to be unique in the following sense. Let $\hat{K}'$ be an extension of K together with an absolute value $|\cdot|'$ extending $|\cdot|$. Assume that $(\hat{K}', |\cdot|')$ is complete and that K is dense in $\hat{K}'$. Then there exists a unique K -isomorphism $\sigma : \hat{K} \rightarrow \hat{K}'$ such that $|\cdot|' \circ \sigma = |\cdot|$.

2.2.2 Completions of archimedean fields

It turns out that complete archimedean valued fields are very easy to understand, as the following Theorem shows:

Theorem 2.2.5

Let $(K, |\cdot|)$ be a complete archimedean valued field. Then there exists an isomorphism σ from K to $\mathbb{R}$ or $\mathbb{C}$ and a real number $s \geq 1$ such that $|\sigma(x)| = |x|^s$ for all $x \in K$.

Proof. By Theorem 2.1.9, the restriction of $|\cdot|$ to $\mathbb{Q}$ is $|\cdot|_\infty^t$ for some $t > 0$. The inequality $|2| \leq 2$ gives $t \leq 1$. Since $\mathbb{R}$ is the completion $\hat{\mathbb{Q}}$ of $\mathbb{Q}$ with respect to $|\cdot|_\infty$ and K is complete, the embedding $\mathbb{Q} \subset K$ extends to an embedding $\tau : \mathbb{R} \subset K$ that identifies $\mathbb{R}$ with the closure of $\mathbb{Q}$ in K .

Fix now $x \in K$ and, for each $z \in \mathbb{C}$, set $Q_z(X) := X^2 - 2\operatorname{Re}(z)X + |z|_\infty^2$ and $f(z) := |Q_z(x)|$. Since $\lim_{|z| \rightarrow \infty} f(z) = +\infty$, the function f attains its minimum m at some $z_0 \in \mathbb{C}$. We can moreover choose z_0 so that its modulus is as large as possible. Suppose $m > 0$ and choose $\epsilon > 0$ with $\epsilon^t < m$. The product of the roots of $Q_{z_0} + \epsilon$ is $|z_0|_\infty^2 + \epsilon$, so one root z_1 has modulus larger than $|z_0|_\infty$, so $f(z_1) > m$. For $n \geq 1$, the monic real polynomial

$$H_n = Q_{z_0}^n - (-\epsilon)^n = \prod_{j=1}^{2n} (X - \alpha_j), \quad \alpha_1 = z_1,$$

satisfies $H_n^2 = \prod_j Q_{\alpha_j}$. Consequently,

$$f(z_1)m^{2n-1} \leq |H_n(x)|^2 \leq (m^n + \epsilon^{tn})^2.$$

Dividing by m^{2n} and letting $n \rightarrow \infty$ contradicts $f(z_1) > m$. Thus $m = 0$ and $Q_{z_0}(x) = 0$. We have therefore proved that every element of K has degree at most two over $\mathbb{R}$. In other words, K is $\mathbb{R}$ or $\mathbb{C}$.

If $K = \mathbb{R} = \hat{\mathbb{Q}}$, there is nothing more to prove. Assume that $K = \mathbb{C}$. For $z = a + bi \in K^\times$, we have $|z| \leq |a|_\infty^t + |b|_\infty^t \leq 2|z|_\infty^t$. Applying this to z^n and making $n \rightarrow \infty$ shows that $|z| \leq |z|_\infty^t$. Applying this last inequality to z^{-1} shows that $|z| \geq |z|_\infty^t$. Hence $|z| = |z|_\infty^t$, and the statement follows with $s = 1/t$. $\square$

2.2.3 Completions of nonarchimedean discretely valued fields

In this section, we focus on the completion of a nonarchimedean discretely valued field.

Proposition 2.2.6

Let $(K, |\cdot|)$ be a nonarchimedean valued field whose associated valuation v is discrete and normalized. Let $\mathcal{O}$ and $\hat{\mathcal{O}}$ be the valuation rings of K and $\hat{K}$ respectively and $\mathfrak{p}$ and $\hat{\mathfrak{p}}$ the corresponding maximal ideals.

1. The valuation on $\hat{K}$ is also discrete and normalized.
2. For every $n \geq 1$ we have a canonical isomorphism:

$$\mathcal{O}/\mathfrak{p}^n \cong \hat{\mathcal{O}}/\hat{\mathfrak{p}}^n.$$

Moreover, the diagonal map:

$$\begin{aligned} \delta : \hat{\mathcal{O}} &\rightarrow \varprojlim_{n \geq 1} \hat{\mathcal{O}}/\hat{\mathfrak{p}}^n := \{(x_n)_n \in \prod_{n \geq 1} \hat{\mathcal{O}}/\hat{\mathfrak{p}}^n : \forall n, x_n = x_{n+1} \bmod \hat{\mathfrak{p}}^n\} \\ a &\mapsto (a \bmod \hat{\mathfrak{p}}, a \bmod \hat{\mathfrak{p}}^2, \dots) \end{aligned}$$

is an isomorphism and a homeomorphism (where each $\hat{\mathcal{O}}/\hat{\mathfrak{p}}^n$ is endowed with the discrete topology and $\varprojlim_n \hat{\mathcal{O}}/\hat{\mathfrak{p}}^n$ is endowed with the projective limit topology).

3. Let $\mathcal{R} \subset \mathcal{O}$ be a system of representatives of the residue field $k := \mathcal{O}/\mathfrak{p}$ containing 0. Let π be a uniformizer of K . Then every element $x \in \hat{\mathcal{O}}$ can be uniquely written as:

$$x = \sum_{n=0}^{\infty} a_n \pi^n$$

with $a_0, a_1, \dots \in \mathcal{R}$.

Proof. 1. For $x \in \hat{K}^\times$, choose $a \in K$ with $|x - a| < |x|$. The ultrametric inequality gives $|a| = |x|$, hence $v(\hat{K}^\times) = v(K^\times) = \mathbb{Z}$.

2. Consider the natural inclusion $\iota: \mathcal{O} \rightarrow \hat{\mathcal{O}}$. We have:

$$\mathfrak{p}^n = \{x \in \mathcal{O} : v(x) \geq n\} = \{x \in \hat{\mathcal{O}} : v(x) \geq n\} \cap \mathcal{O} = \hat{\mathfrak{p}}^n \cap \mathcal{O}.$$

Hence ι induces an injective morphism:

$$\bar{\iota}: \mathcal{O}/\mathfrak{p}^n \rightarrow \hat{\mathcal{O}}/\hat{\mathfrak{p}}^n.$$

For the surjectivity, take some $y \in \hat{\mathcal{O}}$. Since K is dense in $\hat{K}$, we can find a sequence $(y_m)_m$ in K converging to y . Since $\hat{\mathfrak{p}}^n$ is open in $\hat{K}$, for m large enough:

$$y_m \in (y + \hat{\mathfrak{p}}^n) \cap \mathcal{O},$$

so that $\bar{\iota}(y_m) = y$.

We now consider the diagonal map δ . We have:

$$\ker(\delta) = \bigcap_n \hat{\mathfrak{p}}^n = v^{-1}(\infty) = \{0\},$$

so δ is injective. For the surjectivity, take $(\bar{a}_n)_n \in \varprojlim_n \hat{\mathcal{O}}/\hat{\mathfrak{p}}^n$ and for each n let a_n be a lifting of $\bar{a}_n$ in $\mathcal{O}$. For $n, m \geq 1$, we have $v(a_m - a_n) \geq \min\{m, n\}$, hence the sequence $(a_n)_n$ is Cauchy. Let a be its limit in $\hat{K}$. We have $v(a - a_n) \geq n$ for all n , hence $a_n = a \pmod{\hat{\mathfrak{p}}^n}$ and $\delta(a) = (\bar{a}_n)_n$.

The subgroups $\hat{\mathfrak{p}}^m$ form a neighbourhood basis of 0 in $\hat{\mathcal{O}}$. Their images under δ are precisely the kernels of the projections to $\hat{\mathcal{O}}/\hat{\mathfrak{p}}^m$, which form a neighbourhood basis of 0 in the inverse limit. Thus δ is a homeomorphism.

3. We construct inductively a sequence $(a_n)_n$ with values in $\mathcal{R}$ such that $v(x - \sum_{n=0}^m a_n \pi^n) \geq m+1$ for all $m \geq 0$. For $m=0$, it suffices to choose $a_0 \in \mathcal{R}$ so that $a_0 \equiv x \pmod{\hat{\mathfrak{p}}}$. Now assume that $a_0, \dots, a_{m-1}$ have been constructed. We then have:

$$v(x - \sum_{n=0}^{m-1} a_n \pi^n) \geq m.$$

In other words, we can write:

$$x - \sum_{n=0}^{m-1} a_n \pi^n = c_m \pi^m$$

for some $c_m \in \hat{\mathcal{O}}$. Choose $a_m \in \mathcal{R}$ so that $a_m \equiv c_m \pmod{\hat{\mathfrak{p}}}$. Then:

$$v(x - \sum_{n=0}^m a_n \pi^n) = v(\pi^m (c_m - a_m)) \geq m + 1,$$

as wished. We get $x = \sum_{n=0}^{\infty} a_n \pi^n$.

Now take $(b_n)_n$ a sequence with values in $\mathcal{R}$ such that $x = \sum_{n=0}^{\infty} b_n \pi^n$ and assume that $(a_n)_n \neq (b_n)_n$. Let n_0 be the smallest integer such that $a_{n_0} \neq b_{n_0}$. Then:

$$0 = \sum_{n=0}^{\infty} a_n \pi^n - \sum_{n=0}^{\infty} b_n \pi^n = (a_{n_0} - b_{n_0}) \pi^{n_0} + \sum_{n \geq n_0+1} (a_n - b_n) \pi^n,$$

with $v((a_{n_0} - b_{n_0}) \pi^{n_0}) = n_0$ and $v(\sum_{n \geq n_0+1} (a_n - b_n) \pi^n) \geq n_0 + 1$: contradiction! We deduce the uniqueness of the sequence $(a_n)_n$. □

Definition 2.2.7

Given a prime number p , the completion of $(\mathbb{Q}, |\cdot|_{v_p})$ is denoted $\mathbb{Q}_p$ and is called the **field of p -adic numbers**. Its valuation ring is denoted $\mathbb{Z}_p$ and is called the **ring of p -adic integers**. We have $\mathbb{Z}_p / p^n \mathbb{Z}_p \cong \mathbb{Z} / p^n \mathbb{Z}$ for each $n \geq 1$, and $\mathbb{Z}_p \cong \varprojlim_{n \geq 1} \mathbb{Z} / p^n \mathbb{Z}$. By Tychonoff's Theorem, it is a compact ring.

2.2.4 Hensel's Lemma

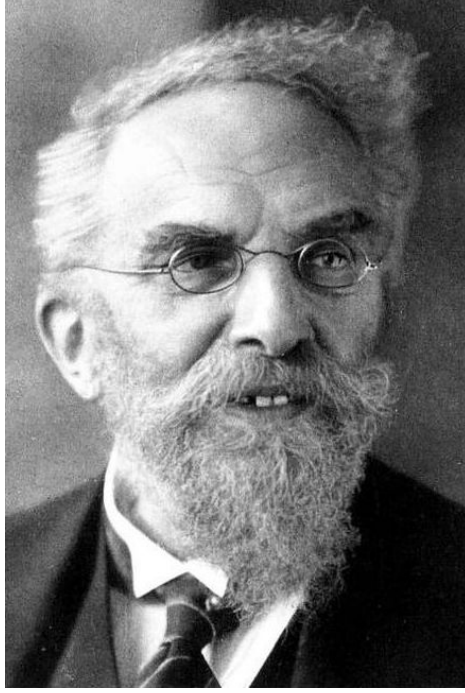
It turns out that, borrowing ideas from analysis (and in particular Newton's approximation), it is quite often relatively easy to find roots of polynomials in complete non-archimedean valued fields. This is one of the reasons why it is usually easier to work with p -adic fields instead of number fields.

Theorem 2.2.8 (Hensel)

Let $(K, |\cdot|)$ be a complete nonarchimedean valued field with valuation ring $\mathcal{O}$. Consider a polynomial $f \in \mathcal{O}[X]$ and an element $a \in \mathcal{O}$ such that $f'(a) \neq 0$ and $c := |f(a)| |f'(a)|^{-2} < 1$. Then the sequence $(a_n)_n$ defined by:

$$\begin{cases} a_0 = a \\ a_{n+1} = a_n - \frac{f(a_n)}{f'(a_n)} \end{cases}$$

is well defined and converges to a root b of f in $\mathcal{O}$ such that $|b - a| \leq c < 1$.



Kurt Hensel (Königsberg 1861 - Marburg 1941)

Proof. It suffices to prove that, for all $n \geq 0$:

$$\begin{aligned} a_n \in \mathcal{O}, \quad |a_{n+1} - a_n| \leq c^{2^n}, \quad |a_n - a| \leq c, \\ |f'(a_n)| = |f'(a)| \neq 0, \quad |f(a_n)| \leq c^{2^n} |f'(a_n)|^2. \end{aligned}$$

Indeed, the inequality $|a_m - a_n| \leq c^{2^n}$ for $m > n$ and the completeness of K imply that the sequence $(a_n)_n$ converges to some $b \in \mathcal{O}$, and the inequalities $|a_n - a| \leq c$ and $|f(a_n)| \leq c^{2^n} |f'(a_n)|^2 = c^{2^n} |f'(a)|^2$ show that $f(b) = 0$ and that $|b - a| \leq c$.

We now prove the claimed properties of the sequence $(a_n)_n$ by induction. The case $n = 0$ is obvious. Now assume that a_n satisfies the desired conditions for some n . Then:

$$|a_{n+1} - a_n| = |f(a_n) f'(a_n)^{-1}| \leq c^{2^n} |f'(a_n)| \leq c^{2^n} < 1,$$

hence $a_{n+1} \in \mathcal{O}$ and:

$$|a_{n+1} - a| \leq \max\{|a_{n+1} - a_n|, |a_n - a|\} \leq c.$$

Now using Taylor's expansions:

$$f'(a_{n+1}) = f'(a_n) + b'_n \frac{f(a_n)}{f'(a_n)},$$

$$f(a_{n+1}) = f(a_n) - \frac{f(a_n)}{f'(a_n)} f'(a_n) + b_n \frac{f(a_n)^2}{f'(a_n)^2} = b_n \frac{f(a_n)^2}{f'(a_n)^2}$$

for some $b_n, b'_n \in \mathcal{O}$. Since $|f(a_n) f'(a_n)^{-1}| < |f'(a_n)|$, we deduce that:

$$|f'(a_{n+1})| = \max\{|f'(a_n)|, |b'_n f(a_n) f'(a_n)^{-1}|\} = |f'(a_n)| = |f'(a)| \neq 0,$$

$$|f(a_{n+1})| \leq |f(a_n) f'(a_n)^{-2}|^2 |f'(a_n)|^2 \leq c^{2^{n+1}} |f'(a_n)|^2 = c^{2^{n+1}} |f'(a_{n+1})|^2.$$

□

Corollary 2.2.9

Let $(K, |\cdot|)$ be a complete nonarchimedean valued field with valuation ring $\mathcal{O}$ and residue field κ . Consider a polynomial $f \in \mathcal{O}[X]$ and assume that the reduction $\bar{f}$ of f in $\kappa[X]$ has a simple root $\bar{a} \in \kappa$. Then f has a unique root a in $\mathcal{O}$ that lifts $\bar{a}$.

Proof. Fix some lift $b \in \mathcal{O}$ of $\bar{a}$. We have $|f(b)| < 1$ and $|f'(b)| = 1$, so existence of a follows from Hensel's lemma. If a' is another root of f in $\mathcal{O}$ lifting $\bar{a}$, then Taylor expansion gives

$$0 = f(a) - f(a') = (a - a')(f'(a') + (a - a')h)$$

for some $h \in \mathcal{O}$. But $f'(a') + (a - a')h \in \mathcal{O}^\times$, so $a = a'$. □

Example 2.2.10. Consider $f = X^3 - 19 \in \mathbb{Z}_3[X]$. We have $v_3(f(-2)) = 3$ and $v_3(f'(-2)) = 1$. By Hensel's lemma, we deduce that 19 has a cubic root in $\mathbb{Z}_3$.

2.2.5 Extensions of complete valued fields

In this subsection, we study finite extensions of a complete valued field K . If L/K is such a finite extension, then L can be seen as a normed K -vector space in the following sense:

Definition 2.2.11

Let $(K, |\cdot|)$ be a valued field. Let V be a finite-dimensional K -vector space. A **norm** on V is a map $\|\cdot\|: V \rightarrow \mathbb{R}_{\geq 0}$ such that:

1. $\forall x \in V, \|x\| = 0 \Leftrightarrow x = 0$;
2. $\forall \lambda \in K, \forall x \in V, \|\lambda x\| = |\lambda| \|x\|$;
3. $\forall x, y \in V, \|x + y\| \leq \|x\| + \|y\|$.

Another norm $\|\cdot\|'$ on V is said to be equivalent to $\|\cdot\|$ if there exist constants $C_1, C_2 > 0$ such that $C_1 \|\cdot\| \leq \|\cdot\|' \leq C_2 \|\cdot\|$.

Proposition 2.2.12

Let $(K, |\cdot|)$ be a complete valued field and let V be a K -vector space with finite dimension n . All norms on V are equivalent. In particular, V is complete and homeomorphic to K^n .

Proof. The case $V = 0$ is immediate. Assume $n \geq 1$. Fix a basis $\mathbf{v} = (v_1, \dots, v_n)$ of V and introduce the norm defined by:

$$\left\| \sum_{i=1}^n \lambda_i v_i \right\|_{\mathbf{v}} = \max\{|\lambda_i| : 1 \leq i \leq n\}.$$

We prove that every norm on V is equivalent to $\|\cdot\|_{\mathbf{v}}$. Indeed, let us fix a norm $\|\cdot\|$ on V . For every $x = \sum_{i=1}^n \lambda_i v_i$, we have:

$$\|x\| \leq \sum_{i=1}^n |\lambda_i| \|v_i\| \leq \left(\sum_{i=1}^n \|v_i\| \right) \|x\|_{\mathbf{v}},$$

hence $\|\cdot\| \leq C_2 \|\cdot\|_{\mathbf{v}}$ with $C_2 = \sum_{i=1}^n \|v_i\|$. Let us now prove by induction on $n = \dim V$ that there exists $C_1 > 0$ such that $C_1 \|\cdot\|_{\mathbf{v}} \leq \|\cdot\|$. If $\dim V = 1$, one can take $C_1 = \|v_1\|$. Assume now that we have already dealt with the case when V has dimension $n-1$, and set $V_j := \bigoplus_{i \neq j} K v_i$ for $1 \leq j \leq n$. By the inductive assumption, we know that each V_j is complete, hence closed in V . We can therefore find $C_1 > 0$ such that $B(0, C_1) \cap \bigcup_{j=1}^n (V_j + v_j) = \emptyset$. Hence, for $x = \sum_{i=1}^n \lambda_i v_i \in V \setminus \{0\}$, we can find $1 \leq k \leq n$ with $|\lambda_k| = \max\{|\lambda_i| : 1 \leq i \leq n\} = \|x\|_{\mathbf{v}}$, so that:

$$\|x\| = |\lambda_k| \left\| \frac{\lambda_1}{\lambda_k} v_1 + \dots + v_k + \dots + \frac{\lambda_n}{\lambda_k} v_n \right\| \geq C_1 \|x\|_{\mathbf{v}},$$

as wished. □

Theorem 2.2.13

Let $(K, |\cdot|)$ be a complete valued field. Let L be a finite extension of K . Then there exists a unique absolute value on L extending the absolute value on K . It is given by $|x|_L = |N_{L/K}(x)|^{1/[L:K]}$ for $x \in L$ and $(L, |\cdot|_L)$ is complete.

Note that the case when K is archimedean immediately follows from Theorem 2.2.5.

Proof when K is a discrete valuation field and L/K is separable. We denote by v the valuation on K . We may and do assume that v is normalized. According to Proposition 2.1.10, the valuation ring $\mathcal{O}$ of K is a principal ideal domain. Hence the integral closure A of $\mathcal{O}$ in L is a Dedekind domain and we can find a non-zero prime ideal $\mathfrak{q}$ of A lying above the maximal ideal $\mathfrak{p}$ of $\mathcal{O}$. Consider the $\mathfrak{p}$ -adic valuation $v_{\mathfrak{p}} : K \rightarrow \mathbb{Z} \cup \{\infty\}$. By definition, we have $v_{\mathfrak{p}}|_K = e(\mathfrak{q}/\mathfrak{p}) \cdot v$, hence $\frac{1}{e(\mathfrak{q}/\mathfrak{p})} v_{\mathfrak{p}}$ is a discrete valuation on L that extends v .

We now want to prove the uniqueness of the extension. So let $|\cdot|_1$ and $|\cdot|_2$ be two absolute values on L extending $|\cdot|$. By Proposition 2.2.12, the absolute values $|\cdot|_1$ and $|\cdot|_2$ are equivalent as norms on the finite-dimensional K -vector space L . In particular, they are equivalent as absolute values. By Proposition 2.1.6, there exists $s > 0$ such that $|\cdot|_1 = |\cdot|_2^s$. But $|\cdot|_1$ and $|\cdot|_2$ coincide on K . Hence $s = 1$ and $|\cdot|_1 = |\cdot|_2$. We henceforth denote by $|\cdot|_L$ the unique absolute value on L that extends $|\cdot|$.

The valued field $(L, |\cdot|_L)$ is complete by Proposition 2.2.12. For each embedding $\sigma \in \text{Hom}_K(L, M)$, the composite $|\cdot|_M \circ \sigma$ is an absolute value on L extending $|\cdot|$. Hence, by uniqueness, $|\cdot|_M \circ \sigma = |\cdot|_L$. We deduce that:

$$|\cdot|_L = (|\cdot|_L^{[L:K]})^{1/[L:K]} = \left(\prod_{\sigma} (|\cdot|_M \circ \sigma) \right)^{1/[L:K]} = (|\cdot|_M \circ N_{L/K})^{1/[L:K]} = (|\cdot| \circ N_{L/K})^{1/[L:K]}.$$

□

Corollary 2.2.14

Let $(K, |\cdot|)$ be a valued field with completion $(\hat{K}, |\cdot|_{\hat{K}})$. Let Ω be an algebraic closure of $\hat{K}$ and let L/K be a finite extension.

1. There exists a unique absolute value $|\cdot|_{\Omega}$ on Ω extending $|\cdot|_{\hat{K}}$.
2. Every absolute value on L extending $|\cdot|$ is of the form $|\cdot|_{\Omega} \circ \sigma$ for some $\sigma \in \text{Hom}_K(L, \Omega)$.

3. Given $\sigma, \tau \in \text{Hom}_K(L, \Omega)$, we have $|\cdot|_\Omega \circ \sigma = |\cdot|_\Omega \circ \tau$ if, and only if, there exists $\rho \in \text{Gal}(\Omega/\hat{K})$ such that $\tau = \rho \circ \sigma$.

Proof. 1. For each finite subextension $M/\hat{K}$ of Ω , there exists a unique absolute value $|\cdot|_M$ on M extending $|\cdot|_{\hat{K}}$. The uniqueness ensures that, if we have a tower of finite extensions $M'/M/\hat{K}$ contained in Ω , then $(|\cdot|_{M'})|_M = |\cdot|_M$. This compatibility allows to define an absolute value $|\cdot|_\Omega$ on Ω extending $|\cdot|$. Moreover, if $|\cdot|'_\Omega$ is another such absolute value, we have $(|\cdot|'_\Omega)|_M = (|\cdot|_\Omega)|_M$ for each finite subextension M of $\Omega/\hat{K}$, hence $|\cdot|'_\Omega = |\cdot|_\Omega$.

2. Let $|\cdot|_L$ be some extension of $|\cdot|$ to L . Let $(\hat{L}, |\cdot|_{\hat{L}})$ be the completion of L . The closure of K in $\hat{L}$ identifies with $\hat{K}$. The compositum $\hat{K}L$ is finite-dimensional over $\hat{K}$, hence complete by Proposition 2.2.12 and closed in $\hat{L}$. It contains the dense subfield L , so $\hat{L} = \hat{K}L$ and $\hat{L}/\hat{K}$ is finite. We may therefore find some embedding $\sigma \in \text{Hom}_{\hat{K}}(\hat{L}, \Omega)$. Both $|\cdot|_{\hat{L}}$ and $|\cdot|_\Omega \circ \sigma$ are absolute values on $\hat{L}$ that extend $|\cdot|_{\hat{K}}$. Hence $|\cdot|_{\hat{L}} = |\cdot|_\Omega \circ \sigma$, and $|\cdot|_L = |\cdot|_\Omega \circ \sigma|_L$.

3. Assume that $\tau = \rho \circ \sigma$ for some $\rho \in \text{Gal}(\Omega/\hat{K})$. By 1, we have $|\cdot|_\Omega \circ \rho = |\cdot|_\Omega$, hence $|\cdot|_\Omega \circ \sigma = |\cdot|_\Omega \circ \tau$.

Conversely, assume $|\cdot|_\Omega \circ \sigma = |\cdot|_\Omega \circ \tau$. Consider the K -isomorphism $\rho_0 := \tau \circ \sigma^{-1} : \sigma(L) \rightarrow \tau(L)$. We claim that it can be extended continuously to a $\hat{K}$ -isomorphism $\hat{\rho}_0 : \sigma(L)\hat{K} \rightarrow \tau(L)\hat{K}$. Indeed, take some $x \in \sigma(L)\hat{K}$ and choose some sequence $(x_n)_n$ with values in $\sigma(L)$ converging to x . For all $m, n \geq 1$, we have $|\rho_0(x_m) - \rho_0(x_n)|_\Omega = |\tau(\sigma^{-1}(x_m - x_n))|_\Omega = |x_m - x_n|_\Omega$, hence the sequence $(\rho_0(x_n))_n$ is Cauchy. Since $\tau(L)\hat{K}$ is complete, it has a limit $\hat{\rho}_0(x)$. One easily checks that this limit does not depend on the choice of the sequence $(x_n)_n$ and that $\hat{\rho}_0$ is a $\hat{K}$ -isomorphism. It can then be extended to a $\hat{K}$ -automorphism $\rho \in \text{Gal}(\Omega/\hat{K})$ and we get $\tau = \rho \circ \sigma$. $\square$

2.3 Ramification theory

2.3.1 Complete discretely valued fields

In this section, we fix $(K, |\cdot|)$ a complete valued field whose associated valuation v is discrete. Let L be a finite separable extension of K . We still denote by $|\cdot|$

and v the extension of absolute value and the valuation of K to L .

Let $\mathcal{O}_K$ and $\mathcal{O}_L$ be the valuation rings of K and L , $\mathfrak{p}_K$ and $\mathfrak{p}_L$ their corresponding maximal ideals and κ_K and κ_L their corresponding residue fields. Introduce the ramification index $e = e(L/K) = e(\mathfrak{p}_L/\mathfrak{p}_K)$ and the residual degree $f = f(L/K) = f(\mathfrak{p}_L/\mathfrak{p}_K)$. The equality $\mathfrak{p}_K \mathcal{O}_L = \mathfrak{p}_L^e$ provides a new interpretation of e as the index of $v(K^\times)$ inside $v(L^\times)$. Also, by Chapter 1, we have the fundamental relation $ef = [L : K]$. Following the definitions of that chapter, we say that L/K is unramified if $e = 1$ and λ/κ is separable, and that it is totally ramified if $f = 1$.

Lemma 2.3.1. *Let L be a finite separable extension of K and take $a \in \mathcal{O}_L$. Suppose that $f(a) = 0$ for a monic polynomial $f \in \mathcal{O}_K[X]$ and that $\bar{a}$ is a simple root of $\bar{f} \in \kappa_K[X]$. Then $K(a)/K$ is unramified and*

$$\kappa_{K(a)} = \kappa_K(\bar{a}), \quad [K(a) : K] = [\kappa_K(\bar{a}) : \kappa_K].$$

Proof. Let $\bar{g}$ be the minimal polynomial of $\bar{a}$ over κ_K , of degree d , and choose a monic lift $g \in \mathcal{O}_K[X]$. Since $\bar{a}$ is separable, Hensel's lemma gives a root $b \in \mathcal{O}_L$ of g reducing to $\bar{a}$. Thus

$$d \leq [\kappa_{K(b)} : \kappa_K] \leq [K(b) : K] \leq d.$$

Hence $K(b)/K$ is unramified with residue field $\kappa_K(\bar{a})$. Hensel's lemma applied to f over $K(b)$ gives a root reducing to $\bar{a}$; such a root is unique because $\bar{a}$ is a simple root of $\bar{f}$ and hence is necessarily equal to a . Consequently $K(a) \subseteq K(b)$ and:

$$d \leq [\kappa_{K(a)} : \kappa_K] \leq [K(a) : K] \leq [K(b) : K] = d.$$

So $K(a) = K(b)$, and the lemma follows. $\square$

Proposition 2.3.2

Let L and L' be two finite unramified extensions of K seen inside some separable closure K^{sep} of K . Then the composite field LL' is unramified over K .

Proof. Let $\bar{a}$ be a primitive element of the extension κ_L/κ_K and choose a lift $a \in \mathcal{O}_L$. Since

$$[L : K] = [\kappa_K(\bar{a}) : \kappa_K] \leq [K(a) : K] \leq [L : K],$$

we have $L = K(a)$, and the minimal polynomial $\mu \in \mathcal{O}_K[X]$ of a reduces to the separable minimal polynomial of $\bar{a}$. Its reduction remains separable over $\kappa_{L'}$. Lemma 2.3.1, applied over L' , therefore shows that $LL' = L'(a)$ is unramified over L' . Since L'/K is unramified, so is LL'/K . $\square$

Definition 2.3.3

Let L be a finite separable extension of K . The composite of all finite unramified subextensions is called the **maximal unramified subextension** of L/K .

Proposition 2.3.4

Let L/K be a finite separable extension with residue field extension κ_L/κ_K and let M be the maximal unramified subextension. Then the residue field κ_M of M is the separable closure of κ_K in κ_L and $v(M^\times) = v(K^\times)$. Moreover, for each intermediary field $\kappa_M/\kappa_N/\kappa_K$, there exists a unique intermediary extension $M/N/K$ with residue field κ_N .

Proof. Since M/K is unramified, $v(M^\times) = v(K^\times)$ and κ_M/κ_K is separable. Given $\bar{a} \in \kappa_L$ separable over κ_K , lift its minimal polynomial to a monic $\mu \in \mathcal{O}_K[X]$. Hensel's lemma gives a root $a \in \mathcal{O}_L$ reducing to $\bar{a}$. By Lemma 2.3.1, the extension $K(a)/K$ is unramified with residue field $\kappa_K(\bar{a})$. Thus $K(a) \subseteq M$ and $\bar{a} \in \kappa_M$, proving the assertion about κ_M .

For an intermediate field κ_N of κ_M/κ_K , choose a primitive element $\bar{a}$. The previous arguments show that there exists $a \in \mathcal{O}_M$ reducing to $\bar{a}$ and that the field $N := K(a)$ has residue field κ_N . If $N' \subseteq M$ has the same residue field, Hensel's lemma shows that N' contains a conjugate b of a lifting $\bar{a}$, and the separability of $\bar{a}$ implies that $b = a$. Both N/K and N'/K are unramified of degree $[\kappa_N : \kappa_K]$, so $N = N'$. $\square$

Definition 2.3.5

Assume that the residue field κ_K of K has characteristic $p > 0$. A finite separable extension L/K is said to be **tamely ramified** if the residue field extension is separable and the ramification index $e(L/K)$ is prime to p . Otherwise, it is said to be **wildly ramified**.

From now on, in this subsection, we assume that the residue characteristic p is positive.

Lemma 2.3.6. *Let L/K be a finite separable extension with maximal unramified subextension M . Let ϖ be a uniformizer of K and let e be a positive integer prime to p such that $\frac{v(\varpi)}{e} \in v(L^\times)$. Then there exists a uniformizer π of M such that L contains an e -th root of π .*

Proof. Let $x \in L$ such that $v(x) = \frac{v(\varpi)}{e}$. There exists a unit $u \in \mathcal{O}_L^\times$ such that $x^e = u\varpi$. But the residue field κ_M is the separable closure of κ_K in κ_L . Hence there exists $u' \in \mathcal{O}_M^\times$, $u'' \in 1 + \mathfrak{p}_L$ and $r \geq 0$ such that $u^{p^r} = u' u''$. Now by Hensel's Lemma, since $e \wedge p = 1$, there exists $a \in \mathcal{O}_L^\times$ such that $u'' = a^e$. Hence $(a^{-1} x^{p^r})^e = u' \varpi^{p^r}$ and L contains an e -th root y of $u' \varpi^{p^r}$. Choose $m \geq 1$ and $t \geq 0$ such that $mp^r = 1 + et$. Then $(y^m / \varpi^t)^e = u'^m \varpi$, so L contains an e -th root of the uniformizer $\pi := u'^m \varpi$ of M . $\square$

Lemma 2.3.7. *Let π be a uniformizer of K , let $e \geq 1$ be an integer prime to p , and let $L = K(x)$ with $x^e = \pi$. Then L/K is totally ramified of degree e .*

Proof. The polynomial $X^e - \pi \in \mathcal{O}_K[X]$ is Eisenstein, hence irreducible. In particular, $[L : K] = e$. Moreover, since $v(x) = \frac{1}{e} v(\pi)$, we have $e \leq [v(L^\times) : v(K^\times)] \leq [L : K] = e$. We deduce that $[v(L^\times) : v(K^\times)] = e$ and hence L/K is totally ramified. $\square$

Proposition 2.3.8

Let L/K be a finite separable extension with maximal unramified subextension M . Then L/K is tamely ramified if, and only if, there exists a uniformizer π of M and a positive integer e coprime to p such that $L = M(\sqrt[e]{\pi})$.

Proof. If L/K is tamely ramified and we set $e := [L : M]$, then e is prime to p and $v(L^\times) = \frac{1}{e} v(K^\times)$. Hence, by Lemma 2.3.6, there exists a uniformizer π of M and an e -th root $\sqrt[e]{\pi}$ of π contained in L . By Lemma 2.3.7, we have $[M(\sqrt[e]{\pi}) : M] = e = [L : M]$, and hence $L = M(\sqrt[e]{\pi})$.

Conversely, if $L = M(\sqrt[e]{\pi})$, Lemma 2.3.7 implies that L/M is totally ramified of degree e . Since $e \wedge p = 1$, we deduce that L/K is tamely ramified. $\square$

Corollary 2.3.9

Let L and L' be two finite tamely ramified extensions of K seen inside some separable closure K^{sep} of K . Then the composite field LL' is tamely ramified over K .

Proof. Let M and M' be the maximal unramified subextensions of L/K and L'/K respectively. By Proposition 2.3.8, write $L = M(\alpha)$ and $L' = M'(\beta)$ with $\alpha^e = \pi$, $\beta^{e'} = \pi'$, where e, e' are prime to p and π, π' are uniformizers of M, M' . Set $F = MM'$, which is unramified over K , and write $\pi' = u\pi$ with $u \in \mathcal{O}_F^\times$.

Choose z in $\bar{K}$ with $z^{e'} = \alpha$ and set $x = \beta/z^e$. Then $z^{ee'} = \pi$ and $x^{e'} = u$. By the Lemma 2.3.1 applied to $X^{e'} - u$, the extension $F(x)/F$ is unramified. Thus π is still a uniformizer of $F(x)$, and Lemma 2.3.7 shows that $F(x, z)/F(x)$ is totally ramified of degree ee' . Hence $F(x, z)/K$ is tame. Since $\alpha = z^{e'}$ and $\beta = xz^e$, it contains LL' , which is therefore tamely ramified as well. $\square$

Definition 2.3.10

Let L be a finite separable extension of K . The composite of all tamely ramified subextensions of L/K is called the **maximal tamely ramified subextension of L over K** .

Proposition 2.3.11

Let L/K be a finite separable extension with residue field extension κ_L/κ_K and let M and N be the maximal unramified and the maximal tamely ramified subextensions of L/K . Then the residue field κ_N of N is the separable closure of κ_K in κ_L and $v(N^\times) = \frac{1}{m}v(K^\times)$ where m is the largest integer coprime to p dividing $[L : M]$.

Proof. The residue field κ_M of M is the separable closure of κ_K in κ_L . Since $\kappa_M \subset \kappa_N \subset \kappa_L$ and κ_N/κ_K is separable, $\kappa_N = \kappa_M$.

We now compute $v(N^\times)$. Since κ_L/κ_M is purely inseparable, $f(L/M)$ is a power of p . The equality $[L : M] = e(L/M)f(L/M)$ then implies that m is the largest integer prime to p dividing $e(L/K)$. By Lemma 2.3.6, there exists a uniformizer π of M and an m -th root $\sqrt[m]{\pi}$ of π contained in L . By Proposition 2.3.8, the field $M(\sqrt[m]{\pi})$ is contained in N , hence using Lemma 2.3.7:

$$\frac{1}{m}v(K^\times) \subset v(M(\sqrt[m]{\pi})^\times) \subset v(N^\times).$$

For the other inclusion, it suffices to notice that $e(N/M)$ is prime to p and divides $e(L/K)$, hence divides m . $\square$

2.3.2 Non-complete discretely valued fields

In this section, we aim at studying finite extensions of valued fields and comparing them to their completions. We start with the following proposition.

Proposition 2.3.12

Let A be a Dedekind domain with fraction field K . Let L be a finite separable extension of K with Galois closure M and let B be the integral closure of A in L . Let $\mathfrak{p}$ be a non-zero prime ideal of A .

1. The group $\text{Gal}(M/K)$ acts transitively on the set of absolute values on M extending $|\cdot|_{v_{\mathfrak{p}}}$.
2. The absolute values of L that extend $|\cdot|_{v_{\mathfrak{p}}}$ are exactly the $|\cdot|_{v_{\mathfrak{q}}}^{1/e(\mathfrak{q}/\mathfrak{p})}$ for $\mathfrak{q}$ a prime ideal of B lying above $\mathfrak{p}$.

Proof. 1. Let Ω be an algebraic closure of the completion of K at $v_{\mathfrak{p}}$. Corollary 2.2.14 describes all extensions to M as $|\cdot|_{\Omega} \circ \sigma$, with $\sigma \in \text{Hom}_K(M, \Omega)$. Since M/K is normal, any two such embeddings have the same image and differ by an element of $\text{Gal}(M/K)$. This proves transitivity.

2. First observe that each absolute value of the form $|\cdot|_{v_{\mathfrak{q}}}^{1/e(\mathfrak{q}/\mathfrak{p})}$ for $\mathfrak{q}$ a prime ideal of B lying above $\mathfrak{p}$ extends $|\cdot|_{v_{\mathfrak{p}}}$. Conversely, let C be the integral closure of A in M . For each prime $\mathfrak{r}$ above $\mathfrak{p}$, the absolute value $|\cdot|_{v_{\mathfrak{r}}}^{1/e(\mathfrak{r}/\mathfrak{p})}$ extends $|\cdot|_{v_{\mathfrak{p}}}$. These absolute values form a nonempty Galois-stable set, so part 1 shows that they are all the extensions to M . Putting $\mathfrak{q} = \mathfrak{r} \cap B$, we have:

$$v_{\mathfrak{r}}|_L = e(\mathfrak{r}/\mathfrak{q})v_{\mathfrak{q}}, \quad e(\mathfrak{r}/\mathfrak{p}) = e(\mathfrak{r}/\mathfrak{q})e(\mathfrak{q}/\mathfrak{p}),$$

and hence the restriction of $|\cdot|_{v_{\mathfrak{r}}}^{1/e(\mathfrak{r}/\mathfrak{p})}$ to L is $|\cdot|_{v_{\mathfrak{q}}}^{1/e(\mathfrak{q}/\mathfrak{p})}$. One concludes by recalling that every extension of $|\cdot|_{v_{\mathfrak{p}}}$ to L extends further to M by Corollary 2.2.14. □

Theorem 2.3.13

Let A be a principal ideal domain with fraction field K . Let L be a finite separable extension of K and let B be the integral closure of A in L . Let $\mathfrak{p}$ be a non-zero prime ideal of A . We denote by $K_{v_{\mathfrak{p}}}$ the completion of K with respect to $v_{\mathfrak{p}}$ and by $A_{v_{\mathfrak{p}}}$ its valuation ring. The $A_{v_{\mathfrak{p}}}$ -algebra morphism:

$$\begin{aligned} \varphi_A : B \otimes_A A_{v_{\mathfrak{p}}} &\rightarrow \prod_{\mathfrak{q}|\mathfrak{p}} B_{v_{\mathfrak{q}}} \\ b \otimes a &\mapsto (ba)_{\mathfrak{q}|\mathfrak{p}} \end{aligned}$$

and the K_{v_p} -algebra morphism:

$$\begin{aligned}\varphi_K : L \otimes_K K_{v_p} &\rightarrow \prod_{q|p} L_{v_q} \\ x \otimes y &\mapsto (xy)_{q|p}\end{aligned}$$

are both isomorphisms.

Proof. Since L/K is finite separable and A is a principal ideal domain, B is a finite free A -module. Hence the functor $B \otimes_A -$ commutes with inverse limits. Writing $\mathfrak{p}B = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_g^{e_g}$ the prime factorization of $\mathfrak{p}B$, we get thanks to Propositions 2.2.6 and 2.3.12 and to the Chinese Remainder Theorem:

$$\begin{aligned}B \otimes_A A_{v_p} &\cong B \otimes_A \varprojlim_n A/\mathfrak{p}^n \cong \varprojlim_n B/(\mathfrak{p}B)^n \\ &\cong \varprojlim_n B/(\mathfrak{p}_1^{n e_1} \dots \mathfrak{p}_g^{n e_g}) \cong \prod_{i=1}^g \varprojlim_n B/\mathfrak{p}_i^{n e_i} \\ &\cong \prod_{i=1}^g B_{v_{\mathfrak{p}_i}}, \\ L \otimes_K K_{v_p} &\cong B \otimes_A K_{v_p} \cong (B \otimes_A A_{v_p}) \otimes_{A_{v_p}} K_{v_p} \\ &\cong \prod_{i=1}^g (B_{v_{\mathfrak{p}_i}} \otimes_{A_{v_p}} K_{v_p}) \cong \prod_{i=1}^g L_{v_{\mathfrak{p}_i}}.\end{aligned}$$

□

In the rest of this section, we fix a finite separable extension L/K and a discrete valuation v on K . We denote by K_v the completion of K with respect to v . When w is a discrete valuation on L , we write $w|v$ to signify that $w|_K = v$.

Corollary 2.3.14

The K_v -algebra morphism:

$$\begin{aligned}\varphi_K : L \otimes_K K_v &\rightarrow \prod_{w|v} L_w \\ x \otimes y &\mapsto (xy)_{w|v}\end{aligned}$$

is an isomorphism. In particular:

$$[L : K] = \sum_{w|v} e(L_w/K_v) f(L_w/K_v).$$

Proof. One may rescale v and assume it is normalized. One then applies Proposition 2.3.12 and Theorem 2.3.13 with A the valuation ring of v in K and $\mathfrak{p}$ its unique maximal ideal. $\square$

Remark 2.3.15. Corollary 2.3.14 recovers, by different means, the Fundamental Relation of Chapter 1.

To finish this section, let us focus on the case where L/K is Galois with group G . As we have seen, G acts transitively on the discrete valuations of L dividing v . In particular, the integers $e(L_w/K_v)$ and $f(L_w/K_v)$ do not depend on the discrete valuation w dividing v . In the sequel, we will write e_v and f_v instead of $e(L_w/K_v)$ and $f(L_w/K_v)$.

Definition 2.3.16

Let w be a discrete valuation on L dividing v . Denote by $\mathcal{O}_L$ the valuation ring associated to w and by $\mathfrak{p}_L$ its maximal ideal.

1. The stabilizer of w in G is called the **decomposition subgroup** of w and denoted $G_w(L/K)$. The **inertia subgroup** of w is defined by:

$$I_w(L/K) := \{\sigma \in G_w : \forall x \in \mathcal{O}_L, \sigma(x) \equiv x \pmod{\mathfrak{p}_L}\},$$

and the **ramification subgroup** by:

$$R_w(L/K) := \{\sigma \in G_w : \forall x \in L^\times, \frac{\sigma(x)}{x} \equiv 1 \pmod{\mathfrak{p}_L}\}.$$

2. The fixed fields corresponding to $G_w(L/K)$, $I_w(L/K)$ and $R_w(L/K)$ are respectively called the **decomposition subfield** of w (denoted M_w), the **inertia subfield** of w (denoted N_w) and the **ramification subfield** of w (denoted P_w).

Proposition 2.3.17

Given a discrete valuation w on L dividing v , we have canonical isomorphisms:

$$G_w(L/K) \cong \text{Gal}(L_w/K_v),$$

$$I_w(L/K) \cong I_w(L_w/K_v), \quad R_w(L/K) \cong R_w(L_w/K_v).$$

Proof. Denote by $\mathcal{O}_L$ the valuation ring associated to w and by $\mathfrak{p}_L$ its maximal ideal. Each $\sigma \in G_w(L/K)$ is an isometry and extends uniquely to a K_v -

automorphism $\hat{\sigma}$ of L_w . Conversely, a K_v -automorphism of L_w preserves L by normality of L/K , and preserves the valuation by uniqueness of its extension from K_v . Restriction and continuous extension are mutual inverses, proving the first isomorphism. Moreover, by Proposition 2.2.6, completion preserves residue fields, so these maps identify the inertia groups. Now, if $\sigma \in R_w(L/K)$ and $x \in L_w^\times$, choose $a \in L$ with $w(x - a) > w(x)$. Then $w(a) = w(x)$ and

$$\hat{\sigma}(x) - x = \hat{\sigma}(x - a) + (\sigma(a) - a) + (a - x)$$

has valuation greater than $w(x)$. Thus $\hat{\sigma} \in R_w(L_w/K_v)$; the converse is immediate. $\square$

Proposition 2.3.18

Let w be a discrete valuation on L dividing v , and assume that the residue field extension κ_L/κ_K is separable.

1. Let w_M be the restriction of w to M_w . Then w is the only discrete valuation of L extending w_M . Moreover, $e(M_w/K) = f(M_w/K) = 1$ and $M_w = L \cap K_v$.
2. The extension κ_L/κ_K is Galois and there is an exact sequence:

$$1 \rightarrow I_w(L/K) \rightarrow G_w(L/K) \rightarrow \text{Gal}(\kappa_L/\kappa_K) \rightarrow 1.$$

Moreover, N_w is the maximal unramified subextension of L/M_w .

3. If $p = \text{char}(\kappa_K) > 0$, the ramification subgroup $R_w(L/K)$ is the unique p -Sylow subgroup of $I_w(L/K)$, and the ramification subfield P_w is the maximal tamely ramified subextension of L/M_w .

Proof. Set $D = G_w(L/K)$, $I = I_w(L/K)$, $R = R_w(L/K)$, $E = L_w$ and $F = K_v$. Since L/K is Galois, $E = FL$ is a splitting field over F , hence is Galois. Proposition 2.3.17 identifies $D = \text{Gal}(E/F)$; applied to L/L^H , it also identifies the completion of L^H with E^H for every $H \subset D$.

1. Proposition 1.2.27 immediately implies the uniqueness of w and the equalities $e(M_w/K) = f(M_w/K) = 1$. Moreover, with Proposition 2.3.17:

$$M_w = L^{G_w(L/K)} = L \cap L_w^{\text{Gal}(L_w/K_v)} = L \cap K_v.$$

2. This immediately follows from Propositions 1.2.28 and 1.2.30.

3. For a uniformizer π of L , the homomorphism

$$\varphi : I_w(L/K) \longrightarrow \kappa_L^\times, \quad \sigma \longmapsto \overline{\sigma(\pi)/\pi}$$

has kernel $R_w(L/K)$: indeed, if $\sigma \in \ker \varphi$ and if $x = u\pi^n \in L^\times$ with $u \in \mathcal{O}_L^\times$, then $\overline{\sigma(x)/x} = \overline{\sigma(u)/u} = 1$, the last equality coming from the fact that $I_w(L/K) = \ker(G_w(L/K) \rightarrow \text{Gal}(\kappa_L/\kappa_K))$. The finite image of φ has order prime to p , since a field of characteristic p has no nontrivial p -th roots of unity. Moreover, if $R_w(L/K)$ had an element σ with prime order $\ell \neq p$, then for every $x \in L_w^\times$, we would have:

$$y = \frac{1}{\ell} \sum_{j=0}^{\ell-1} \sigma^j(x) \in L_w^{\langle \sigma \rangle}, \quad w(y-x) > w(x).$$

Hence $L_w^{\langle \sigma \rangle}$ would have the same value group and residue field as E , and $[L_w : L_w^{\langle \sigma \rangle}]$ would be 1: contradiction. We deduce that $R_w(L/K)$ has no elements of prime order different from p , hence $R_w(L/K)$ is a p -group. It is a normal p -Sylow subgroup of $I_w(L/K)$, hence the unique p -Sylow subgroup of $I_w(L/K)$. This also immediately implies that P_w is the maximal tamely ramified subextension of L/M_w .

□

2.4 Places of number fields

We finish the chapter by coming back to number fields. We start by classifying places over them:

Theorem 2.4.1

Let K be a number field and let Σ be a system of representatives of $\text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$ under complex conjugation. A system of representatives of the places of K is given by the absolute values $|\cdot|_\sigma$ for $\sigma \in \Sigma$ and by the absolute values $|\cdot|_{v_p}$ for p non-zero prime ideal in $\mathcal{O}_K$.

We start with the following lemma:

Lemma 2.4.2. *Let K be a number field and let $|\cdot|$ be a (non-trivial) absolute value on K . The restriction of $|\cdot|$ to $\mathbb{Q}$ is non-trivial.*

Proof. By contradiction, assume that the restriction of $|\cdot|$ to $\mathbb{Q}$ is trivial. Set $d := [K : \mathbb{Q}]$ and take $x \in K$. Then x satisfies an equation:

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$$

with $n \leq d$ and $a_0, \dots, a_{n-1} \in \mathbb{Q}$. Hence:

$$|x|^n = |a_{n-1}x^{n-1} + \dots + a_0| \leq |x|^{n-1} + \dots + |x| + 1 \leq d \max\{1, |x|\}^{n-1},$$

so that $|x| \leq d$. Applying this argument to x^k with $k \geq 1$, we get $|x|^k \leq d$ for all k . Hence $|\cdot| \leq 1$ and $|\cdot|$ is trivial: contradiction! $\square$

Proof of Theorem 2.4.1. Choose some absolute value $|\cdot|$ on K . Denote by $|\cdot|_{\mathbb{Q}}$ be its restriction to $\mathbb{Q}$. It is not trivial by the previous lemma. If $|\cdot|$ is archimedean, then so is $|\cdot|_{\mathbb{Q}}$ and hence there exists $0 < s \leq 1$ such that $|\cdot|_{\mathbb{Q}} = |\cdot|_{\infty}^s$. By Corollary 2.2.14, denoting by $|\cdot|_{\mathbb{C}}$ the usual complex module, there exists $\sigma \in \text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$ such that $|\cdot| = |\cdot|_{\mathbb{C}}^s \circ \sigma = |\cdot|_{\sigma}^s$, and hence $|\cdot|$ is equivalent to $|\cdot|_{\sigma}$. If $|\cdot|$ is nonarchimedean, then $|\cdot|_{\mathbb{Q}} = |\cdot|_{v_p}^s$ for some prime number p and some $s > 0$. Applying Proposition 2.3.12 to $|\cdot|^{1/s}$ then implies that $|\cdot|$ is equivalent to $|\cdot|_{v_p}$ for some prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$.

Assume now that σ and τ are two embeddings in $\text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$ such that $|\cdot|_{\sigma}$ and $|\cdot|_{\tau}$ are equivalent. Then there exists $s > 0$ such that $|\cdot|_{\sigma} = |\cdot|_{\tau}^s$. By restricting to $\mathbb{Q}$, we get $|\cdot|_{\infty} = |\cdot|_{\infty}^s$ and hence $s = 1$. Moreover, we have $|\cdot|_{\sigma} = |\cdot|_{\bar{\sigma}}$ for each $\sigma \in \text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$, and if $|\cdot|_{\sigma} = |\cdot|_{\tau}$, Corollary 2.2.14(3) gives $\tau = \sigma$ or $\tau = \bar{\sigma}$.

Finally, if $\mathfrak{p}$ and $\mathfrak{q}$ are distinct nonzero prime ideals in $\mathcal{O}_K$, then any element of $\mathfrak{p} \setminus \mathfrak{q}$ has absolute value less than one at $\mathfrak{p}$ and equal to one at $\mathfrak{q}$, hence $|\cdot|_{v_p}$ and $|\cdot|_{v_q}$ are not equivalent. $\square$

Given a number field K , we denote by Ω_K (resp. $\Omega_K^f, \Omega_K^\infty$) the set of places (resp. non-archimedean places, archimedean places) of K . One often studies the arithmetic properties of K by trying to reduce them to the arithmetic properties of the completions of K . This means one sees K as a diagonally embedded subring of $\prod_{v \in \Omega_K} K_v$. This last ring is however often too big, and that is why one introduces an intermediary ring, the so-called ring of adèles. We start with the general notion of restricted product:

Definition 2.4.3

Let $(X_i)_{i \in I}$ be a family of topological spaces and suppose we are given an open subset Y_i of X_i for each i . The **restricted product** of the X_i with respect to the Y_i is defined as:

$$\prod'_{i \in I} X_i := \{(x_i)_{i \in I} \in \prod_{i \in I} X_i : x_i \in Y_i \text{ for almost all } i \in I\} = \bigcup_{S \subset I \text{ finite}} X_S$$

with $X_S := \prod_{i \in S} X_i \times \prod_{i \in I \setminus S} Y_i$. We endow each X_S with the product topol-

ogy and $\prod'_{i \in I} X_i$ with the inductive limit topology.

In particular, given $(x_i)_{i \in I}$, a basis of open neighborhoods of $(x_i)_{i \in I}$ is given by products $\prod_{i \in I} U_i$ where each U_i is open in X_i and contains x_i , and $U_i = Y_i$ for almost all i .

Let us now apply this construction to our arithmetic setting:

Definition 2.4.4

For each $v \in \Omega_K^f$, let $\mathcal{O}_v$ be the valuation ring of K_v . The **ring of adèles** of K , denoted $\mathbb{A}_K$, is the restricted product of the K_v for $v \in \Omega_K$ with respect to the $\mathcal{O}_v$ at finite places and to K_v at archimedean places. In other words, $\mathbb{A}_K = \bigcup_{\Omega_K^\infty \subset S \subset \Omega_K \text{ finite}} \mathbb{A}_{K,S}$ with:

$$\mathbb{A}_{K,S} := \prod_{v \in S} K_v \times \prod_{v \notin S} \mathcal{O}_v.$$

Since an element of K is integral at almost all finite places, the diagonal map embeds K in $\mathbb{A}_K$. The local inclusions $\mathbb{Q}_p \hookrightarrow K_v$ for $v|p$ similarly define an embedding $\mathbb{A}_\mathbb{Q} \hookrightarrow \mathbb{A}_K$.

Proposition 2.4.5

The natural map

$$K \otimes_\mathbb{Q} \mathbb{A}_\mathbb{Q} \longrightarrow \mathbb{A}_K, \quad a \otimes (x_p)_p \longmapsto (ax_p)_{v|p},$$

is an isomorphism of topological K -algebras. Here the topology on the left is obtained by identifying it with $\mathbb{A}_\mathbb{Q}^{[K:\mathbb{Q}]}$ through any $\mathbb{Q}$ -basis of K .

Proof. Theorem 2.3.13 gives

$$\mathcal{O}_K \otimes_\mathbb{Z} \mathbb{Z}_p \cong \prod_{v|p} \mathcal{O}_v, \quad K \otimes_\mathbb{Q} \mathbb{Q}_p \cong \prod_{v|p} K_v.$$

At infinity:

$$K \otimes_\mathbb{Q} \mathbb{R} \cong \mathbb{R}^r \times \mathbb{C}^s \cong \prod_{v \in \Omega_K^\infty} K_v,$$

where $r = |\text{Hom}_\mathbb{Q}(K, \mathbb{R})|$ and $s = \frac{1}{2} (|\text{Hom}_\mathbb{Q}(K, \mathbb{C})| - |\text{Hom}_\mathbb{Q}(K, \mathbb{R})|)$. All the previous isomorphisms are homeomorphisms. Using the fact that $\mathcal{O}_K$ is a finitely

generated free $\mathbb{Z}$ -module, we deduce that, for each finite subset $S \subset \Omega_{\mathbb{Q}}$ containing the infinite place, there is an isomorphism of topological rings

$$\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{A}_{\mathbb{Q},S} \cong \mathbb{A}_{K,S'},$$

where S' is the set of places of K above S . Taking the union of all previous isomorphisms when S varies yields the desired isomorphism $K \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}} \cong \mathbb{A}_K$. $\square$

Corollary 2.4.6

The ring $\mathbb{A}_K$ is locally compact, the diagonal subgroup K is discrete, and $\mathbb{A}_K/K$ is compact. More precisely, there is a compact subset $C \subset \mathbb{A}_K$ such that $\mathbb{A}_K = K + C$.

Proof. By Proposition 2.4.5, it suffices to treat $K = \mathbb{Q}$. The set $C := [-1, 1] \times \prod_p \mathbb{Z}_p$ is a compact neighbourhood of zero, and $\mathbb{Q} \cap ((-1, 1) \times \prod_p \mathbb{Z}_p) = \{0\}$, so $\mathbb{Q}$ is discrete in $\mathbb{A}_{\mathbb{Q}}$. For each prime number p , we have $\mathbb{Q}_p = \mathbb{Z}[1/p] + \mathbb{Z}_p$. Hence, given an adèle $a \in \mathbb{A}_{\mathbb{Q}}$, we can find an integer N and an element $z \in \mathbb{Z}[1/N]$ such that $a - z \in \mathbb{A}_{\mathbb{Q}, \Omega_{\mathbb{Q}}^{\infty}}$. We can then find $z' \in \mathbb{Z}$ such that $a - z - z' \in C$. Thus $\mathbb{A}_{\mathbb{Q}} = \mathbb{Q} + C$. $\square$

We now introduce the unit group of $\mathbb{A}_K$:

Definition 2.4.7

The **group of idèles** of K is the unit group

$$\mathbb{A}_K^{\times} = \left\{ (x_v)_v \in \prod_{v \in \Omega_K} K_v^{\times} : x_v \in \mathcal{O}_v^{\times} \text{ for almost all } v \in \Omega_K^f \right\}.$$

Its topology is the restricted product topology with respect to the groups $\mathcal{O}_v^{\times}$: for each finite $S \supset \Omega_K^{\infty}$, the subgroup

$$\mathbb{A}_{K,S}^{\times} := \prod_{v \in S} K_v^{\times} \times \prod_{v \notin S} \mathcal{O}_v^{\times}$$

is open and has the product topology. The diagonal image of $K^{\times}$ is called the group of **principal idèles**, and $\mathbb{A}_K^{\times}/K^{\times}$ is the **idèle class group**, endowed with the quotient topology.

With the above topology, $\mathbb{A}_K^{\times}$ is a locally compact topological group: the subgroups $\mathbb{A}_{K,S}^{\times}$ are topological groups, and the groups $\mathcal{O}_v^{\times}$ are compact and open

in $K_v^\times$. Its inclusion in $\mathbb{A}_K$ is continuous, so $K^\times$ is discrete in $\mathbb{A}_K^\times$.

However, one has to be careful: the idèle topology is strictly finer than the topology induced by $\mathbb{A}_K$. For example, choose a sequence $(v_j)_{j \geq 0}$ of distinct finite places and, for each j , a uniformizer $\pi_j \in K_{v_j}$. The idèle with component π_j at v_j and 1 elsewhere converges to 1 in $\mathbb{A}_K$, but never belongs to the idèle neighbourhood

$$\prod_{v \in \Omega_K^\infty} K_v^\times \times \prod_{v \in \Omega_K^f} \mathcal{O}_v^\times.$$

In number theory, it is often convenient to work with a normalized version of the absolute values $|\cdot|_v$ for $v \in \Omega_K$. When $v = v_p \in \Omega_K^f$, we define the associated **normalized absolute value** as:

$$\|x\|_v := N(\mathfrak{p})^{-v_p(x)}, \quad \text{for } x \in K_v.$$

When $v = v_\sigma \in \Omega_K^\infty$, we set:

$$\|x\|_v := \begin{cases} |x|_{v_\sigma} & \text{if } K_v = \mathbb{R}, \\ |x|_{v_\sigma}^2 & \text{if } K_v = \mathbb{C}, \end{cases} \quad \text{for } x \in K_v.$$

Be careful: when $K_v = \mathbb{C}$, the map $\|\cdot\|_v$ is not an absolute value.

Proposition 2.4.8 (Product formula)

The map

$$\|\cdot\| : \mathbb{A}_K^\times \longrightarrow \mathbb{R}_{>0}, \quad (x_v)_v \longmapsto \prod_{v \in \Omega_K} \|x_v\|_v$$

is a continuous surjective group homomorphism. For every $a \in K^\times$,

$$\prod_{v \in \Omega_K} \|a\|_v = 1.$$

Proof. On each open subgroup $\mathbb{A}_{K,S}^\times$ the map is a finite product of continuous homomorphisms, so it is continuous and multiplicative. Varying a single archimedean component proves surjectivity. For $a \in K^\times$:

$$\prod_{v \in \Omega_K^\infty} \|a\|_v = |N_{K/\mathbb{Q}}(a)|, \quad \prod_{v \in \Omega_K^f} \|a\|_v = N(a\mathcal{O}_K)^{-1} = |N_{K/\mathbb{Q}}(a)|^{-1}.$$

The product formula follows. $\square$

In particular, this homomorphism descends to a continuous surjection $\mathbb{A}_K^\times / K^\times \rightarrow \mathbb{R}_{>0}$, so the idèle class group is not compact. However, one may introduce the kernel $\mathbb{A}_K^{\times 0} := \ker \|\cdot\|$ and the quotient $C_K^0 := \mathbb{A}_K^{\times 0} / K^\times$, and one may then prove:

Theorem 2.4.9

The group C_K^0 is compact.

Proof. To an idèle x associate the fractional ideal

$$J(x) := \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(x_{\mathfrak{p}})}.$$

By the Class Theorem 1.3.17, we may choose finitely many idèles $t_1, \dots, t_h$ whose ideals represent the ideal classes. By adjusting one archimedean component, we may arrange that $\|t_i\| = 1$ for each i . Given $x \in \mathbb{A}_K^{\times 0}$, we can choose i and $a \in K^{\times}$ with $J(x) = (a)J(t_i)$. Then $y = x/(at_i) \in \mathbb{A}_{K, \Omega_K^{\infty}}^{\times}$ and $\prod_{v|\infty} \|y_v\|_v = 1$.

Put $H := \{(z_v)_v \in \prod_{v \in \Omega_K^{\infty}} K_v : \sum_v z_v = 0\}$. By Proposition 1.3.19 (the key proposition to prove Dirichlet's unit Theorem), the vectors $(\log \|u\|_v)_{v \in \Omega_K^{\infty}}$, for $u \in \mathcal{O}_K^{\times}$, form a complete lattice in H . Hence, if $P \subset H$ is a compact fundamental parallelepiped. One can find $b \in \mathcal{O}_K^{\times}$ such that:

$$(\log \|y/b\|_v)_{v \in \Omega_K^{\infty}} \in P.$$

Thus:

$$x \in K^{\times} \bigcup_{i=1}^h t_i \left(B_{\infty} \times \prod_{v \in \Omega_K^f} \mathcal{O}_v^{\times} \right),$$

where:

$$B_{\infty} := \{(t_v)_v \in \prod_{v \in \Omega_K^{\infty}} K_v^{\times} : (\log \|t_v\|_v)_{v|\infty} \in P\}.$$

Hence:

$$C_K^0 = \bigcup_{i=1}^h t_i \left(B_{\infty} \times \prod_{v \in \Omega_K^f} \mathcal{O}_v^{\times} \right).$$

But B_{∞} and $\prod_{v \in \Omega_K^f} \mathcal{O}_v^{\times}$ are both compact, hence so is C_K^0 . □

Remark 2.4.10. This proof uses the Class Theorem and Dirichlet's Unit Theorem. One can also go the other way around, prove directly the compactness of C_K^0 and then deduce the Class Theorem and Dirichlet's Unit Theorem.

We finish this Chapter by proving an approximation Theorem that applies only to number fields but which is in that context much stronger than Artin-Whaples weak approximation Theorem:

Theorem 2.4.11 (Strong approximation)

Let S be any proper subset of Ω_K , possibly infinite, and let $\mathbb{A}_{K,S}^{\text{res}} := \prod'_{v \in S} K_v$ be the restricted product with respect to $\mathcal{O}_v$ at finite places and K_v at archimedean places. Then the diagonal image of K in $\mathbb{A}_{K,S}^{\text{res}}$ is dense.

Proof. Choose a compact set C with $\mathbb{A}_K = K + C$, as in Corollary 2.4.6. The open sets $\mathbb{A}_{K,T}$, for finite $T \supset \Omega_K^\infty$, form a directed cover of $\mathbb{A}_K$. Since C is compact, it is contained in one of them and its coordinate projections are compact, hence bounded. Thus there are numbers $\delta_v > 0$, equal to 1 almost everywhere, such that

$$C \subset \prod_v \{x \in K_v : \|x\|_v \leq \delta_v\}.$$

Let $a \in \mathbb{A}_{K,S}^{\text{res}}$ and let U be an open neighbourhood of a . Then U contains a set of the form:

$$X_T := \{(x_v)_{v \in S} \in \mathbb{A}_K^{\text{res}} : \forall v \in T, \|x_v - a_v\|_v < \epsilon_v; \forall v \in S \setminus T, x_v \in \mathcal{O}_v\},$$

where $T \subset S$ is finite, contains $S \cap \Omega_K^\infty$, and contains every finite place where a_v is not integral. Choose positive numbers r_v for $v \in \Omega_K$ so that:

$$\begin{aligned} \forall v \in T, r_v \delta_v < \epsilon_v, \quad \forall v \in S \setminus T, r_v \delta_v \leq 1, \quad \forall v \in \Omega_K^f, r_v \in \|K_v^\times\|_v, \\ \prod_{v \in \Omega_K} r_v > \left(\frac{2}{\pi}\right)^s \sqrt{|\Delta_K|}, \quad r_v = 1 \text{ for almost all } v, \end{aligned}$$

where s stands for the number of complex embeddings of K modulo conjugation. This can always be done because $S \neq \Omega_K$. For $v = v_p \in \Omega_K^f$, write $r_v = N(\mathfrak{p})^{-n_p}$ and set $I = \prod_p \mathfrak{p}^{n_p}$. We then have $\prod_{v \in \Omega_K^f} r_v = N(I)^{-1}$, and hence $\prod_{v \in \Omega_K^\infty} r_v > \left(\frac{2}{\pi}\right)^s N(I) \sqrt{|\Delta_K|}$. By Theorem 1.3.10, we deduce that there exists $b \in I \setminus \{0\}$ such that $\|b\|_v \leq r_v$ for all $v \in \Omega_K^\infty$.

Now extend a by zero outside S to an adèle $\tilde{a} \in \mathbb{A}_K$. Since

$$\mathbb{A}_K = b\mathbb{A}_K = K + bC,$$

write $\tilde{a} = k + bc$ with $k \in K$ and $c \in C$. At places of T we have $\|a_v - k\|_v < \epsilon_v$. At places of $S \setminus T$, both a_v and bc_v are integral, hence so is k . Thus $k \in U$, and K is dense in $\mathbb{A}_{K,S}^{\text{res}}$. $\square$